Course 5042 | Semester V | Theory | 4 Credits

Advanced Communication Systems

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5042 as Advanced Communication Systems in Semester V for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kanpur’s Principles of Communication Systems course and polytechnic communication curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Network designs, spectrum allocations, link budgets and equipment specifications must be based on approved engineering plans, certified hardware data, current regulatory standards and competent professional review.

Verified source note: Verified sources support sampling, quantization, PCM, digital modulation (PSK, QAM), optical fiber and satellite communication. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Advanced Communication Systems studies the technologies that form the backbone of modern global connectivity. It develops the ability to analyze digital communication techniques (PCM, DPCM, DM), evaluate digital modulation schemes (PSK, QAM), understand optical fiber and satellite communication links, and coordinate wireless network standards (GSM, LTE, 5G) while respecting the technical and regulatory boundaries of telecommunication engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□□ □□□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Analog communication, signals and systems, digital electronics, electromagnetic wave theory and engineering mathematics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain digital communication fundamentals, including sampling, quantization and pulse code modulation (PCM).
2. Explain digital modulation and demodulation techniques for efficient data transmission over various channels.
3. Explain the principles, components and performance of optical fiber and satellite communication systems.
4. Explain the architecture and standards of modern wireless and mobile communication networks.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain digital pulse modulation techniques and digital codes for data representation.	Understand
CO2	Explain digital carrier modulation schemes and their performance in the presence of noise.	Understand
CO3	Explain the working principles of optical fiber and satellite communication links.	Understand
CO4	Explain the architecture and standards of mobile communication and wireless networks.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Digital Pulse Modulation and Coding	Sampling theorem and aliasing; quantization (uniform and non-uniform); Pulse Code Modulation (PCM); Differential PCM (DPCM); Delta Modulation (DM); Adaptive Delta Modulation (ADM); digital codes (line coding); TDM.	Approx. 16 hours	CO1	Module 1
2	Digital Carrier Modulation Techniques	Amplitude Shift Keying (ASK); Frequency Shift Keying (FSK); Phase Shift Keying (PSK); Binary PSK (BPSK); Quadrature PSK (QPSK); Quadrature Amplitude Modulation (QAM); M-ary modulation; Bit Error Rate (BER) basics.	Approx. 16 hours	CO2	Module 2
3	Optical and Satellite Communication	Optical fiber types (SMF, MMF); light sources (LED, Laser); detectors (PIN, APD); losses and dispersion; Satellite orbits (LEO, MEO, GEO); transponders; earth station; link budget; multiple access (FDMA, TDMA, CDMA).	Approx. 14 hours	CO3	Module 3
4	Wireless and Mobile Communication	Evolution of mobile generations (1G to 5G); GSM architecture; GPRS; CDMA; LTE and VoLTE; 5G features and small cells; Wi-Fi (802.11); Bluetooth; IoT communication protocols (LoRa, NB-IoT).	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Digital Pulse Modulation and Coding

Sampling theorem and aliasing; quantization (uniform and non-uniform); Pulse Code Modulation (PCM); Differential PCM (DPCM); Delta Modulation (DM); Adaptive Delta Modulation (ADM); digital codes (line coding); TDM.

CO1 · Approx. 16 hours

Digital Carrier Modulation Techniques

Amplitude Shift Keying (ASK); Frequency Shift Keying (FSK); Phase Shift Keying (PSK); Binary PSK (BPSK); Quadrature PSK (QPSK); Quadrature Amplitude Modulation (QAM); M-ary modulation; Bit Error Rate (BER) basics.

CO2 · Approx. 16 hours

Optical and Satellite Communication

Optical fiber types (SMF, MMF); light sources (LED, Laser); detectors (PIN, APD); losses and dispersion; Satellite orbits (LEO, MEO, GEO); transponders; earth station; link budget; multiple access (FDMA, TDMA, CDMA).

CO3 · Approx. 14 hours

Wireless and Mobile Communication

Evolution of mobile generations (1G to 5G); GSM architecture; GPRS; CDMA; LTE and VoLTE; 5G features and small cells; Wi-Fi (802.11); Bluetooth; IoT communication protocols (LoRa, NB-IoT).

CO4 · Approx. 14 hours

MODULE 1

Digital Pulse Modulation and Coding

Sampling theorem and aliasing; quantization (uniform and non-uniform); Pulse Code Modulation (PCM); Differential PCM (DPCM); Delta Modulation (DM); Adaptive Delta Modulation (ADM); digital codes (line coding); TDM.

Approx. 16 hours **CO1** **Understand · Apply**

Learning goals

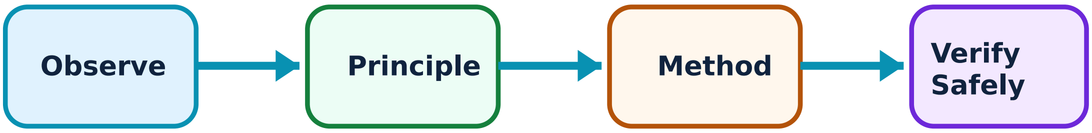
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Digital Pulse Modulation and Coding: identify the system, state its principle, follow the correct method and verify the result safely.

1. PCM and delta modulation−

Digital communication starts with converting analog signals to digital. PCM involves sampling, quantization and coding. DPCM and Delta Modulation reduce bandwidth by transmitting only the difference between successive samples. Adaptive Delta Modulation (ADM) further improves performance by varying the step size to prevent slope overload.

Study and application method

State the 'Sampling Theorem' and explain the effect of aliasing; compare PCM and Delta Modulation in a conceptual table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State the 'Sampling Theorem' and explain the effect of aliasing; compare PCM and Delta Modulation in a conceptual table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Quantization introduces noise that cannot be removed. Do not design communication links without authorised signal-to-quantization-noise ratio (SQNR) analysis.

2. Line coding and multiplexing−

Line coding (e.g., NRZ, RZ, Manchester) adapts digital data for transmission over a channel, ensuring synchronization and bandwidth efficiency. Time Division Multiplexing (TDM) allows multiple users to share a single channel by allocating specific time slots.

Study and application method

Sketch the Manchester and NRZ-L line codes for a given bit sequence; explain the principle of T1 carrier system in TDM.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the Manchester and NRZ-L line codes for a given bit sequence; explain the principle of T1 carrier system in TDM.

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Common mistake / precaution: Multiplexing requires precise timing and synchronization. Ensure all clock recovery circuits follow authorised engineering standards to prevent data loss.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Digital Carrier Modulation Techniques

Amplitude Shift Keying (ASK); Frequency Shift Keying (FSK); Phase Shift Keying (PSK); Binary PSK (BPSK); Quadrature PSK (QPSK); Quadrature Amplitude Modulation (QAM); M-ary modulation; Bit Error Rate (BER) basics.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

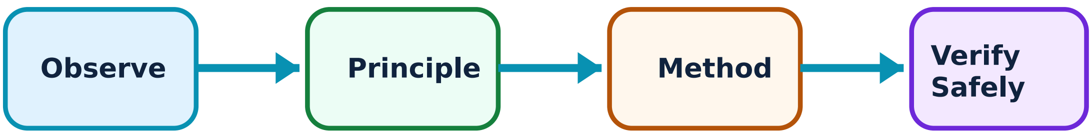
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Digital Carrier Modulation Techniques: identify the system, state its principle, follow the correct method and verify the result safely.

1. Digital modulation schemes−

Digital modulation shifts a carrier's amplitude, frequency or phase to represent digital data. BPSK and QPSK are efficient phase-based schemes. QAM combines amplitude and phase modulation to achieve higher data rates within a given bandwidth, essential for modern Wi-Fi and cellular networks.

Study and application method

Sketch the constellation diagram for BPSK and QPSK; explain why QAM is more bandwidth-efficient than BPSK.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the constellation diagram for BPSK and QPSK; explain why QAM is more bandwidth-efficient than BPSK.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High-order modulation (e.g., 64-QAM) is sensitive to noise and phase jitter. Do not use high-order schemes without authorised link-budget and signal-integrity verification.

2. Performance and noise−

The performance of modulation schemes is evaluated using Bit Error Rate (BER) vs Signal-to-Noise Ratio (SNR). Optimum receivers (Matched Filter) are designed to minimize BER in the presence of Additive White Gaussian Noise (AWGN).

Study and application method

Define 'Bit Error Rate' (BER); explain the concept of a 'Matched Filter' in a digital receiver conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Bit Error Rate' (BER); explain the concept of a 'Matched Filter' in a digital receiver conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Noise levels vary with environment and hardware. All receiver designs must meet authorised sensitivity and interference-rejection standards.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Optical and Satellite Communication

Optical fiber types (SMF, MMF); light sources (LED, Laser); detectors (PIN, APD); losses and dispersion; Satellite orbits (LEO, MEO, GEO); transponders; earth station; link budget; multiple access (FDMA, TDMA, CDMA).

Approx. 14 hours | **CO3** | **Understand · Apply**

Learning goals

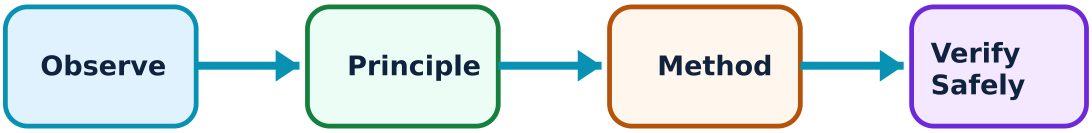
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Optical and Satellite Communication: identify the system, state its principle, follow the correct method and verify the result safely.

1. Optical fiber links−

Optical fibers transmit data as light pulses. Single-mode fibers (SMF) provide higher bandwidth for long distances. Light sources like Lasers and detectors like Avalanche Photodiodes (APD) are used. Dispersion and attenuation limit the distance and data rate of the link.

Study and application method

Compare Single-mode and Multi-mode fibers; list the factors contributing to attenuation in an optical fiber.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare Single-mode and Multi-mode fibers; list the factors contributing to attenuation in an optical fiber.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Optical fibers and lasers involve radiation hazards. Do not look directly into an energized fiber or laser source without authorised eye protection.

2. Satellite communication systems−

Satellites act as repeaters in space. Geostationary (GEO) satellites are common for broadcasting. The link budget accounts for all gains and losses from the earth station to the satellite and back. Multiple access techniques allow many users to share the satellite transponder.

Study and application method

Explain the role of a 'Transponder' in a satellite; describe the advantages of FDMA and TDMA in satellite communication.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the role of a 'Transponder' in a satellite; describe the advantages of FDMA and TDMA in satellite communication.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Satellite links involve high latency and complex frequency coordination. All satellite operations must follow authorised ITU spectrum allocations and orbital slots.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Wireless and Mobile Communication

Evolution of mobile generations (1G to 5G); GSM architecture; GPRS; CDMA; LTE and VoLTE; 5G features and small cells; Wi-Fi (802.11); Bluetooth; IoT communication protocols (LoRa, NB-IoT).

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

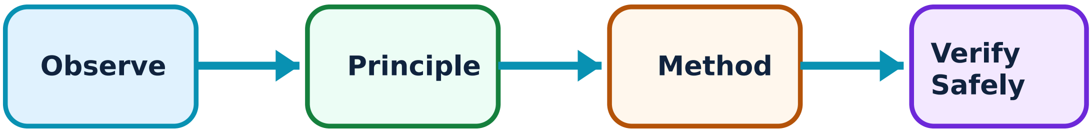
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Wireless and Mobile Communication: identify the system, state its principle, follow the correct method and verify the result safely.

1. Mobile network architecture−

Mobile networks use a cellular structure to reuse frequencies. GSM introduced digital voice and SMS. LTE (4G) moved to an all-IP packet-switched architecture for high-speed data. 5G offers ultra-low latency and massive connectivity using millimeter waves and small cells.

Study and application method

Sketch the basic architecture of a GSM network; list three key features of 5G technology.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the basic architecture of a GSM network; list three key features of 5G technology.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Wireless networks produce electromagnetic radiation. All base station installations must meet authorised safety limits for RF exposure (e.g., ICNIRP guidelines).

2. Wireless standards and IoT−

Wi-Fi and Bluetooth provide short-range connectivity. For the Internet of Things (IoT), low-power wide-area networks (LPWAN) like LoRa and NB-IoT are used to connect thousands of sensors over long distances with minimal power consumption.

Study and application method

Compare Wi-Fi and Bluetooth in terms of range and data rate; explain the need for LPWAN in IoT applications.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare Wi-Fi and Bluetooth in terms of range and data rate; explain the need for LPWAN in IoT applications.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Unlicensed wireless bands (e.g., 2.4GHz) are prone to interference. Ensure all wireless deployments follow authorised frequency usage and coexistence protocols.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Digital Pulse Modulation and Coding

Use the concept flow to revise observation, principle, method and verification.

Module 2: Digital Carrier Modulation Techniques

Use the concept flow to revise observation, principle, method and verification.

Module 3: Optical and Satellite Communication

Use the concept flow to revise observation, principle, method and verification.

Module 4: Wireless and Mobile Communication

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare ASK, FSK and PSK in a conceptual table showing their bandwidth and noise immunity.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the block diagram of a PCM transmitter and receiver.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the structure of a typical optical fiber and label the core, cladding and coating.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the satellite link budget conceptually given the transmit power, antenna gains and path loss.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the mobile generation (e.g., 4G/5G) and frequency band currently used by your smartphone.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Digital Pulse Modulation and Coding

Explain PCM and delta modulation and state one correct method, application or precaution.−

Digital communication starts with converting analog signals to digital. PCM involves sampling, quantization and coding. DPCM and Delta Modulation reduce bandwidth by transmitting only the difference between successive samples. Adaptive Delta Modulation (ADM) further improves performance by varying the step size to prevent slope overload.

Answer framework: State the 'Sampling Theorem' and explain the effect of aliasing; compare PCM and Delta Modulation in a conceptual table.

Important point: Quantization introduces noise that cannot be removed. Do not design communication links without authorised signal-to-quantization-noise ratio (SQNR) analysis.

Explain Line coding and multiplexing and state one correct method, application or precaution.−

Line coding (e.g., NRZ, RZ, Manchester) adapts digital data for transmission over a channel, ensuring synchronization and bandwidth efficiency. Time Division Multiplexing (TDM) allows multiple users to share a single channel by allocating specific time slots.

Answer framework: Sketch the Manchester and NRZ-L line codes for a given bit sequence; explain the principle of T1 carrier system in TDM.

Important point: Multiplexing requires precise timing and synchronization. Ensure all clock recovery circuits follow authorised engineering standards to prevent data loss.

Module 2 — Digital Carrier Modulation Techniques

Explain Digital modulation schemes and state one correct method, application or precaution.−

Digital modulation shifts a carrier's amplitude, frequency or phase to represent digital data. BPSK and QPSK are efficient phase-based schemes. QAM combines amplitude and phase modulation to achieve higher data rates within a given bandwidth, essential for modern Wi-Fi and cellular networks.

Answer framework: Sketch the constellation diagram for BPSK and QPSK; explain why QAM is more bandwidth-efficient than BPSK.

Important point: High-order modulation (e.g., 64-QAM) is sensitive to noise and phase jitter. Do not use high-order schemes without authorised link-budget and signal-integrity verification.

Explain Performance and noise and state one correct method, application or precaution.−

The performance of modulation schemes is evaluated using Bit Error Rate (BER) vs Signal-to-Noise Ratio (SNR). Optimum receivers (Matched Filter) are designed to minimize BER in the presence of Additive White Gaussian Noise (AWGN).

Answer framework: Define 'Bit Error Rate' (BER); explain the concept of a 'Matched Filter' in a digital receiver conceptually.

Important point: Noise levels vary with environment and hardware. All receiver designs must meet authorised sensitivity and interference-rejection standards.

Module 3 — Optical and Satellite Communication

Explain Optical fiber links and state one correct method, application or precaution.−

Optical fibers transmit data as light pulses. Single-mode fibers (SMF) provide higher bandwidth for long distances. Light sources like Lasers and detectors like Avalanche Photodiodes (APD) are used. Dispersion and attenuation limit the distance and data rate of the link.

Answer framework: Compare Single-mode and Multi-mode fibers; list the factors contributing to attenuation in an optical fiber.

Important point: Optical fibers and lasers involve radiation hazards. Do not look directly into an energized fiber or laser source without authorised eye protection.

Explain Satellite communication systems and state one correct method, application or precaution.−

Satellites act as repeaters in space. Geostationary (GEO) satellites are common for broadcasting. The link budget accounts for all gains and losses from the earth station to the satellite and back. Multiple access techniques allow many users to share the satellite transponder.

Answer framework: Explain the role of a 'Transponder' in a satellite; describe the advantages of FDMA and TDMA in satellite communication.

Important point: Satellite links involve high latency and complex frequency coordination. All satellite operations must follow authorised ITU spectrum allocations and orbital slots.

Module 4 — Wireless and Mobile Communication

Explain Mobile network architecture and state one correct method, application or precaution.−

Mobile networks use a cellular structure to reuse frequencies. GSM introduced digital voice and SMS. LTE (4G) moved to an all-IP packet-switched architecture for high-speed data. 5G offers ultra-low latency and massive connectivity using millimeter waves and small cells.

Answer framework: Sketch the basic architecture of a GSM network; list three key features of 5G technology.

Important point: Wireless networks produce electromagnetic radiation. All base station installations must meet authorised safety limits for RF exposure (e.g., ICNIRP guidelines).

Explain Wireless standards and IoT and state one correct method, application or precaution.−

Wi-Fi and Bluetooth provide short-range connectivity. For the Internet of Things (IoT), low-power wide-area networks (LPWAN) like LoRa and NB-IoT are used to connect thousands of sensors over long distances with minimal power consumption.

Answer framework: Compare Wi-Fi and Bluetooth in terms of range and data rate; explain the need for LPWAN in IoT applications.

Important point: Unlicensed wireless bands (e.g., 2.4GHz) are prone to interference. Ensure all wireless deployments follow authorised frequency usage and coexistence protocols.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Digital Pulse Modulation and Coding.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Digital Carrier Modulation Techniques.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Optical and Satellite Communication.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Wireless and Mobile Communication.

8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Sampling theorem and aliasing; quantization (uniform and non-uniform); Pulse Code Modulation (PCM); Differential PCM (DPCM); Delta Modulation (DM); Adaptive Delta Modulation (ADM); digital codes (line coding); TDM..
2. Develop a complete answer or practical plan for: Amplitude Shift Keying (ASK); Frequency Shift Keying (FSK); Phase Shift Keying (PSK); Binary PSK (BPSK); Quadrature PSK (QPSK); Quadrature Amplitude Modulation (QAM); M-ary modulation; Bit Error Rate (BER) basics..
3. Develop a complete answer or practical plan for: Optical fiber types (SMF, MMF); light sources (LED, Laser); detectors (PIN, APD); losses and dispersion; Satellite orbits (LEO, MEO, GEO); transponders; earth station; link budget; multiple access (FDMA, TDMA, CDMA)..
4. Develop a complete answer or practical plan for: Evolution of mobile generations (1G to 5G); GSM architecture; GPRS; CDMA; LTE and VoLTE; 5G features and small cells; Wi-Fi (802.11); Bluetooth; IoT communication protocols (LoRa, NB-IoT)..

LAST-MILE REVISION

Quick Revision

Module 1: Digital Pulse Modulation and Coding

Sampling theorem and aliasing; quantization (uniform and non-uniform); Pulse Code Modulation (PCM); Differential PCM (DPCM); Delta Modulation (DM); Adaptive Delta Modulation (ADM); digital codes (line coding); TDM.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Digital Carrier Modulation Techniques

Amplitude Shift Keying (ASK); Frequency Shift Keying (FSK); Phase Shift Keying (PSK); Binary PSK (BPSK); Quadrature PSK (QPSK); Quadrature Amplitude Modulation (QAM); M-ary modulation; Bit Error Rate (BER) basics.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Optical and Satellite Communication

Optical fiber types (SMF, MMF); light sources (LED, Laser); detectors (PIN, APD); losses and dispersion; Satellite orbits (LEO, MEO, GEO); transponders; earth station; link budget; multiple access (FDMA, TDMA, CDMA).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Wireless and Mobile Communication

Evolution of mobile generations (1G to 5G); GSM architecture; GPRS; CDMA; LTE and VoLTE; 5G features and small cells; Wi-Fi (802.11); Bluetooth; IoT communication protocols (LoRa, NB-IoT).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
PCM and delta modulation	Digital communication starts with converting analog signals to digital.
Line coding and multiplexing	Line coding (e.
Digital modulation schemes	Digital modulation shifts a carrier's amplitude, frequency or phase to represent digital data.
Performance and noise	The performance of modulation schemes is evaluated using Bit Error Rate (BER) vs Signal-to-Noise Ratio (SNR).
Optical fiber links	Optical fibers transmit data as light pulses.
Satellite communication systems	Satellites act as repeaters in space.
Mobile network architecture	Mobile networks use a cellular structure to reuse frequencies.
Wireless standards and IoT	Wi-Fi and Bluetooth provide short-range connectivity.

LEARNING RESOURCES

References

Recommended communication-systems references

1. B. P. Lathi, Modern Digital and Analog Communication Systems.
2. Simon Haykin, Communication Systems.
3. Gerd Keiser, Optical Fiber Communications.
4. Theodore S. Rappaport, Wireless Communications: Principles and Practice.

Approved public communication-systems sources

- https://nptel.ac.in/courses/108104091
- https://onlinecourses.nptel.ac.in/noc21_ee11/preview

These sources provide the verified study basis while official Course 5042 detail is unavailable; they do not replace official network designs, authorised spectrum plans, certified equipment specs or authorised telecom protocols.